(a) Feature 33

Precision 1.00 Recall 0.60

TARGET seller revenue

DIRECTION increase (+0.691 generation rho)

EXPLANATION

The feature tends to activate in revenue-rich auctions, where meaningful competition generates substantial allocations and payments, rather than in low-revenue or no-sale regimes.

TRUE POSITIVE Prediction: active

v0 = [0.28, 0.40, 0.50, 0.71, 0.37]
v1 = [0.19, 0.21, 0.20, 0.05, 0.52]
A0 = [0.77, 0.95, 0.98, 0.99, 0.51]
A1 = [0.21, 0.05, 0.02, 0.00, 0.49]
Payments [b0, b1] = [1.44, 0.29]

Revenue = 1.725 (percentile 94.5)

TRUE NEGATIVE Prediction: inactive

v0 = [0.05, 0.16, 0.07, 0.08, 0.34]
v1 = [0.12, 0.09, 0.09, 0.08, 0.02]
A0 = [0.00, 0.00, 0.00, 0.00, 0.00]
A1 = [0.00, 0.00, 0.00, 0.00, 0.00]
Payments [b0, b1] = [0.00, 0.00]

Revenue = 0.000 (percentile 0.3)

(b) Feature 1

Precision 0.35 Recall 0.36

TARGET bidder utility

DIRECTION decrease (-0.342 generation rho)

EXPLANATION

The feature activates when the focal bidder-item is in a low-truthful-utility regime, typically because the bidder receives little allocation or pays away most of its allocated value.

FALSE POSITIVE Prediction: active

v0 = [0.20, 0.05, 0.08, 0.02, 0.22]
v1 = [0.18, 0.26, 0.41, 0.42, 0.19]
A0 = [0.00, 0.00, 0.00, 0.00, 0.00]
A1 = [0.98, 0.96, 0.99, 1.00, 0.98]
Payments [b0, b1] = [0.00, 0.94]

Utility (b0) = 0.000 (percentile 11.5)

FALSE NEGATIVE Prediction: inactive

v0 = [0.23, 0.65, 0.34, 0.43, 0.34]
v1 = [0.25, 0.02, 0.01, 0.16, 0.06]
A0 = [0.97, 1.00, 1.00, 1.00, 1.00]
A1 = [0.01, 0.00, 0.00, 0.00, 0.00]
Payments [b0, b1] = [1.06, 0.00]

Utility (b0) = 0.934 (percentile 99.3)